Maximum Subarray using DNC

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Course: Design Analysis and Algorithm
Code: BCSE204P
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Code:

```
#include <iostream>
#include <vector>
#include <algorithm>
#include <climits>

using namespace std;

struct SubarrayData {
    int sum;
    int start;
    int end;
};

SubarrayData maxCrossingSum(vector<int>& nums, int l, int m, int h) {
    int sum = 0;
    int left_sum = INT_MIN;
    int max_left = m;
    for (int i = m; i >= l; i--) {
        sum += nums[i];
        if (sum > left_sum) {
            left_sum = sum;
            max_left = i;
        }
    }
    sum = 0;
    int right_sum = INT_MIN;
    int max_right = m + 1;
    for (int i = m + 1; i <= h; i++) {
        sum += nums[i];
        if (sum > right_sum) {
            right_sum = sum;
            max_right = i;
        }
    }
    return {left_sum + right_sum, max_left, max_right};
}

SubarrayData maxSubArraySum(vector<int>& nums, int l, int h) {
    if (l == h) return {nums[l], l, h};
    int m = l + (h - l) / 2;

    SubarrayData left = maxSubArraySum(nums, l, m);
```

```
    SubarrayData right = maxSubArraySum(nums, m + 1, h);
    SubarrayData cross = maxCrossingSum(nums, l, m, h);

    if (left.sum >= right.sum && left.sum >= cross.sum) return left;
    if (right.sum >= left.sum && right.sum >= cross.sum) return right;
    return cross;
}

int main() {
    cout << "Name: Mistry Nishk Nileshkumar\nRegistration Number: 24BCE5236\n";
    int n;
    cout << "Enter size: "; cin >> n;
    vector<int> nums(n);
    cout << "Enter elements: ";
    for (int i = 0; i < n; i++) cin >> nums[i];

    SubarrayData result = maxSubArraySum(nums, 0, n - 1);

    cout << "Maximum Sum: " << result.sum << endl;
    cout << "The Subarray is: ";
    for (int i = result.start; i <= result.end; i++) {
        cout << nums[i] << " ";
    }
    cout << endl;
    return 0;
}
```

Input & Output:

```
Activities   Terminal                                   Jan 13  16:13
student@admin205-ThinkCentre-neo-50s-Gen-5: ~/Documents/24BCE5236
student@admin205-ThinkCentre-neo-50s-Gen-5:~$ cd Documents/
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents$ ls
23BDS1106            24BCE5236        24brs1007lab4.html       'ARYAN(25BYB1173).docx'          DAA.dotx     labassignment6.docx          NoMachine        program     style.css   ub.html
'23BRS1406 FAT LAB.pdf'   24brs1007.html  'ARYAN(25BYB1173)1.docx'   codetantra-sea_4.3.0_amd64.deb   index.html   Neo-Browser-1.1.8.AppImage   portfoll.html   program.c   ubbb.css    WEBPROGRAMMINGLAB5.docx
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents$ cd 24BCE5236
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ls
Maximum_subarray.cpp
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray.cpp -o Maximum_subarray
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
6student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray.cpp -o Maximum_subarray
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
6
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray.cpp -o Maximum_subarray
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Maximum Sum:6
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray.cpp -o Maximum_subarray
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Maximum Sum:6
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray.cpp -o Maximum_subarray
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Enter the number of elements in the array: 5
Enter 5 integers: 7 -2 1 2 -8
Maximum Sum: 8
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ g++ Maximum_subarray1.cpp -o Maximum_subarray1
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$ ./Maximum_subarray1
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Enter size: 5
Enter elements: 7 -2 1 2 -8
Maximum Sum: 8
The Subarray is: 7 -2 1 2
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/24BCE5236$
```